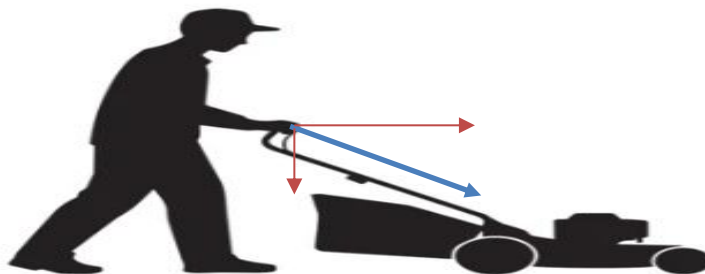


1.3 Dot Products & Projections

Motivation: Suppose you are pushing a lawn mower (like in the picture below). To push the lawn mower you must apply some amount of force (represented by the blue vector, we will call $\mathbf{b}$), however some of this force actually contributes to pushing the lawn mower forward (the horizontal red vector, we will call $\mathbf{a}$) while the rest contributes to pushing the lawn mower directly into the ground (the vertical red vector) which contributes nothing to the motion of the mower.



How can we determine “how much” of vector $\mathbf{b}$ “goes in the direction of” vector $\mathbf{a}$? To answer this question, we need a new operation for vectors.

Definition: If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then the dot product of $\mathbf{a}$ and $\mathbf{b}$ is the number $\mathbf{a} \cdot \mathbf{b}$ given by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Note: The dot product is defined similarly in V_n when $n \geq 2$.

Note: It isn't obvious, but the dot product operation can help us determine the angle between two vectors and answer the question of “how much” of a vector $\mathbf{b}$ “goes in the direction” of a vector $\mathbf{a}$.

Theorem: Suppose $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in V_3 and c is a scalar, then

1. $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$
2. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
3. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
4. $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$
5. $\vec{0} \cdot \mathbf{a} = 0$

Proof of property 1 (you are encouraged to try the others):

Since $\mathbf{a} \in V_3$ we know that $\mathbf{a} = \langle a, b, c \rangle$ for some real numbers a , b , and c . We calculate that

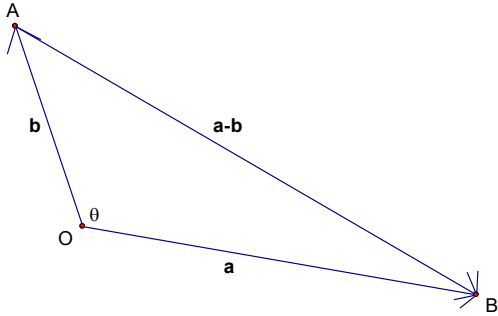
$$\mathbf{a} \cdot \mathbf{a} = \langle a, b, c \rangle \cdot \langle a, b, c \rangle = a^2 + b^2 + c^2 = \left(\sqrt{a^2 + b^2 + c^2} \right)^2 = \|\mathbf{a}\|^2.$$

Note: These properties hold in n dimensions as well.



Theorem: If θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$, then $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$.

Proof (if time permits): First, a sketch of the situation is:



By the law of cosines we know that

$\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$ (crucially, we notice that this formula still holds when $\theta = 0$, $\theta = \pi$, $\mathbf{a} = \mathbf{0}$, or $\mathbf{b} = \mathbf{0}$).

Now, by properties one, two, and three of the previous theorem we have:

$$\begin{aligned} \|\mathbf{a} - \mathbf{b}\|^2 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = (\mathbf{a} - \mathbf{b}) \cdot \mathbf{a} - (\mathbf{a} - \mathbf{b}) \cdot \mathbf{b} = \mathbf{a} \cdot (\mathbf{a} - \mathbf{b}) - \mathbf{b} \cdot (\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b} = \|\mathbf{a}\|^2 - \mathbf{a} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\mathbf{a} \cdot \mathbf{b}. \end{aligned}$$

If we combine this result with what we obtained from the law of cosines we have:

$\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta) \Rightarrow -2\mathbf{a} \cdot \mathbf{b} = -2\|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta) \Rightarrow \mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$. This completes the proof. ■

Note: From the above theorem we immediately obtain the fact that for nonzero vectors $\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$.

Note: Amazingly, in the above theorem we made no assumption about how many components the vectors have! So, theoretically speaking, we can use the above theorem to compute the angle between two vectors in higher than three dimensions!

Note: In some texts the above theorem is taken as a definition (after one proves the Cauchy-Schwarz inequality). This approach is more rigorous since it does not use the assumption that the law of cosines holds in arbitrarily many dimensions (in fact this approach makes no use of the law of cosines at all).

Example 1: Suppose that $\mathbf{a} = \langle 2, 2, -1 \rangle$, $\mathbf{b} = \langle 5, -3, 2 \rangle$, and $\mathbf{c} = \left\langle \sqrt{3}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

a) Find the angle (in radians rounded to the nearest hundredth) between the vectors $\mathbf{a}$ and $\mathbf{b}$.

If θ is the desired angle measure, we know that $\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{10 - 6 - 2}{\sqrt{9} \cdot \sqrt{38}} = \frac{2}{3\sqrt{38}}$. Thus,

$$\theta = \cos^{-1}\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46.$$

b) Find the angle (exactly in radians) between the vectors $\mathbf{c}$ and $\mathbf{i}$ (Note: This angle is known as one of the *directional angles* of the vector $\mathbf{c}$).

If θ is the desired angle measure, we know that $\cos(\theta) = \frac{\mathbf{c} \cdot \mathbf{i}}{\|\mathbf{c}\| \|\mathbf{i}\|} = \frac{\sqrt{3}}{\sqrt{3 + .5 + .5} \cdot \sqrt{1}} = \frac{\sqrt{3}}{2}$. Thus,

$$\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}.$$

c) If the dot product of two nonzero vectors is zero, what can you say about the angle formed between those two vectors?

If θ is the angle measure between the two nonzero vectors, then we know that $\cos(\theta) = 0$ which implies that $\theta = \frac{\pi}{2} = 90^\circ$ and the two vectors are perpendicular.

Definition: Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (one also could say perpendicular) if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.

Note: Under this definition the zero vector is considered orthogonal to all vectors.

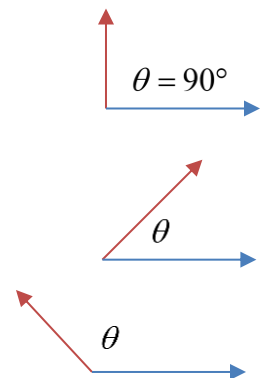
Based on the fact that $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$ we can now establish each of the following cases for understanding what the dot product value tells us about the angle between two vectors.

Implication of the dot product behavior value:

(1) If $\mathbf{a} \cdot \mathbf{b} = 0$ then it must be the case that $\cos(\theta) = 0$ and this implies that $\theta = 90^\circ$.

(2) If $\mathbf{a} \cdot \mathbf{b} > 0$ then $\cos(\theta) > 0$ which implies θ is an acute angle.

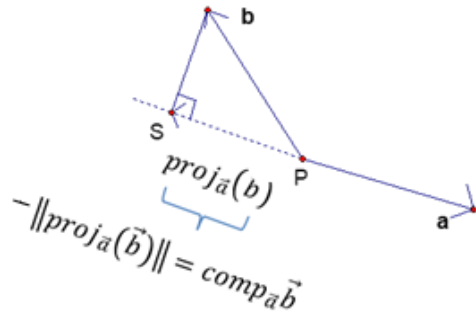
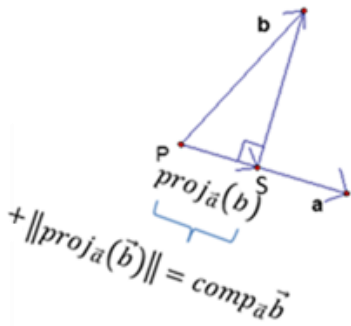
(3) If $\mathbf{a} \cdot \mathbf{b} < 0$ then $\cos(\theta) < 0$ which implies θ is an obtuse angle.



We now return to the important question that we opened the section with. How can we determine “how much” a vector $\mathbf{b}$ is “going in the direction” of a vector $\mathbf{a}$? To answer this question, we first establish two important definitions.

Definitions: Given two vectors $\mathbf{a}$ and $\mathbf{b}$ as shown below

- The *amount of vector $\mathbf{b}$ in the direction of vector $\mathbf{a}$* is called the component of $\mathbf{b}$ onto $\mathbf{a}$ (sometimes called the *scalar projection of $\mathbf{b}$ onto $\mathbf{a}$*) and is written as $\text{comp}_{\mathbf{a}}(\mathbf{b})$. This is represented as the length of the vector $\overrightarrow{PS}$.
- The vector $\overrightarrow{PS}$ is called the projection of $\mathbf{b}$ onto $\mathbf{a}$ (sometimes called the *vector projection of $\mathbf{b}$ onto $\mathbf{a}$*) and is written as $\text{proj}_{\mathbf{a}}(\mathbf{b})$.



Note: The $\text{proj}_{\mathbf{a}}(\mathbf{b})$ is a vector while $\text{comp}_{\mathbf{a}}(\mathbf{b})$ is a scalar.

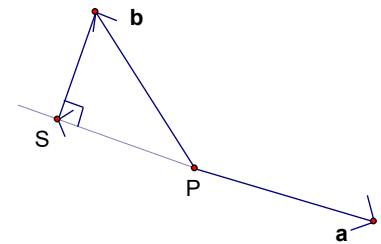
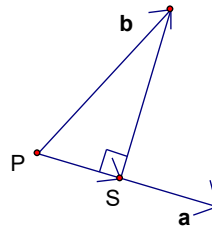
Exploration:

We now wish to derive formulas for both $\text{comp}_{\mathbf{a}}(\mathbf{b})$ and $\text{proj}_{\mathbf{a}}(\mathbf{b})$. We use the following pictures for reference. We refer to the left picture as case 1 and the right picture as case 2.

Suppose that θ is the angle formed between $\mathbf{a}$ and $\mathbf{b}$. Using simple right triangle trigonometry.

Case 1: First, we let $\mathbf{u}$ be a unit vector pointing in the direction of $\mathbf{a}$. Then we calculate that:

$$\text{comp}_{\mathbf{a}}(\mathbf{b}) = \|\overrightarrow{PS}\| = \|\mathbf{b}\| \cos(\theta) = \|\mathbf{u}\| \|\mathbf{b}\| \cos(\theta) = \mathbf{u} \cdot \mathbf{b} = \frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$$



Case 2: First, we let $\mathbf{u}$ be a unit vector pointing in the direction of $\mathbf{a}$. Then we calculate that:

$$\begin{aligned} \text{comp}_{\mathbf{a}}(\mathbf{b}) &= -\|\overrightarrow{PS}\| = -\|\mathbf{b}\| \cos(\pi - \theta) = -\|\mathbf{b}\| [\cos(\pi) \cos(\theta) + \sin(\pi) \sin(\theta)] = \|\mathbf{b}\| \cos(\theta) \\ &= \|\mathbf{u}\| \|\mathbf{b}\| \cos(\theta) = \mathbf{u} \cdot \mathbf{b} = \frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} \end{aligned}$$

So, in any case $\text{comp}_{\mathbf{a}}(\mathbf{b}) = \mathbf{u} \cdot \mathbf{b} = \frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$.

Now, $\text{proj}_{\mathbf{a}}(\mathbf{b})$ is the unit vector in the direction of $\mathbf{a}$ multiplied by the scalar $\text{comp}_{\mathbf{a}}(\mathbf{b})$ (regardless of which

case we are in). So, we have that $\text{proj}_{\mathbf{a}}(\mathbf{b}) = \text{comp}_{\mathbf{a}}(\mathbf{b}) \left(\frac{\mathbf{a}}{\|\mathbf{a}\|} \right) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} \right) \left(\frac{\mathbf{a}}{\|\mathbf{a}\|} \right) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a}$.

Theorem: For vector $\mathbf{b}$ and nonzero vector $\mathbf{a}$ we have the following formulas: $\text{comp}_{\mathbf{a}}(\mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$ and

$$\text{proj}_{\mathbf{a}}(\mathbf{b}) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} \right) \left(\frac{\mathbf{a}}{\|\mathbf{a}\|} \right) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a}.$$

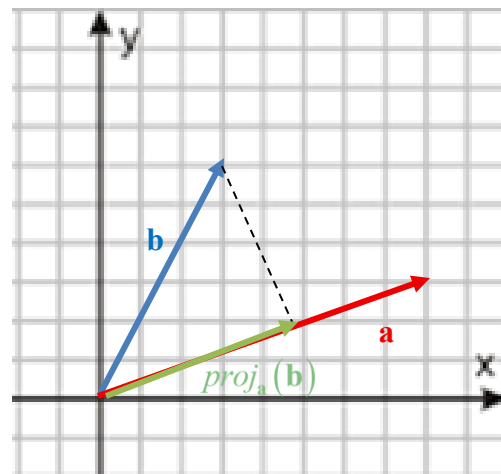
Note: The proof of this theorem is done in the exploration.

Example 2: Find the component and projection of vector $\mathbf{b} = \langle 3, 6 \rangle$ onto $\mathbf{a} = \langle 8, 3 \rangle$. Draw all three vectors on the coordinate axes below starting each vector from $(0, 0)$.

We calculate

$$\text{comp}_{\mathbf{a}}(\mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} = \frac{24 + 18}{\sqrt{9 + 36}} = \frac{42}{\sqrt{45}}$$

$$\text{proj}_{\mathbf{a}}(\mathbf{b}) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = \frac{42}{45} \langle 8, 3 \rangle = \left\langle \frac{336}{45}, \frac{126}{45} \right\rangle$$



Example 3: Find the component and projection of $\mathbf{u} = \langle 1, 1, 2 \rangle$ onto $\mathbf{v} = \langle -2, 3, 1 \rangle$.

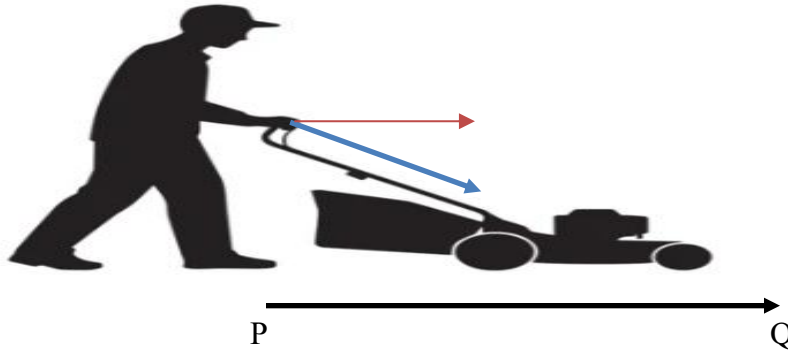
Note that $\mathbf{u} \cdot \mathbf{v} = -2 + 3 + 2 = 3$ and $\|\mathbf{v}\| = \sqrt{4 + 9 + 1} = \sqrt{14}$. Therefore,

$$\text{comp}_{\mathbf{a}}(\mathbf{b}) = \frac{3}{\sqrt{14}}$$

$$\text{proj}_{\mathbf{a}}(\mathbf{b}) = \left(\frac{3}{14} \right) \langle -2, 3, 1 \rangle = \left\langle \frac{-3}{7}, \frac{9}{14}, \frac{3}{14} \right\rangle.$$

Work

Suppose that a constant force vector $\mathbf{F}$ in 3 space moves a particle in a straight line from point P to point Q . The vector $\mathbf{D} = \overrightarrow{PQ}$ is called the *displacement vector* and the *work* done by this force is defined to be the product of the component of the force along $\mathbf{D}$ and the distance moved. Find a simple formula for the work done by $\mathbf{F}$.



By the definition given for work above we know that $W = \text{Force} \cdot \text{Distance}$, however, we must determine the distance moved and how much of the given work vector $\mathbf{F}$ contributes to motion in this direction.

Therefore, we have that $\text{Force} = \text{comp}_{\mathbf{D}}(\mathbf{F})$ and $\text{Distance} = \|\mathbf{D}\|$.

So, this allows us to compute that $W = \text{comp}_{\mathbf{D}}(\mathbf{F})\|\mathbf{D}\| = \frac{\mathbf{F} \cdot \mathbf{D}}{\|\mathbf{D}\|}\|\mathbf{D}\| = \mathbf{F} \cdot \mathbf{D}$

Definition: The amount of **work** performed by a force, $\mathbf{F}$, acting on an object moving in a direction parallel to the force is given by the formula:

$$W = \mathbf{F} \cdot \mathbf{D}$$

Note: Things become much more interesting when the force is variable and the path taken by the particle is complicated. This problem will be tackled much later in MTH 202.